

Jade Chen

🇺🇸 US Citizen 📍 Cypress, TX ✉ astrojade2016@gmail.com ☎ 346 715 7635 in Jade Chen 🐶 Doge2000

Education

University of Texas at Austin

Bachelor of Science: Aerospace Engineering

Aug 2026 – May 2030

Experience

Aerodynamics Engineer

Austin, TX

Texas Flight

September 2026 - Present

- Analyze airfoils in Flow5 at $10,000 < Re < 1.2$ million to support wing-section selection for the AIAA Design/Build/Fly competition aircraft.
- Compared lift, drag, pressure distributions, and pitching moments using Python-based post-processing of Flow5 results to evaluate aerodynamic efficiency, supporting design decisions that improved projected mission endurance.
- Completed composite layups using fiberglass, carbon fiber, epoxy mixing, Mylar application, and vacuum bagging to produce aircraft components.

Software Engineer Intern

Cypress, TX

A+ Computer Science

May 2026 - August 2026

- Diagnosed and rewrote the IDE's error-output handling to convert repetitive, garbled compiler exceptions into concise line-referenced messages, reducing student debugging time by 10-15 minutes.
- Updated the breadcrumb navigation feature in JavaScript, letting users jump directly back to prior problem categories, improving efficiency by 5%
- Hardened account security by enforcing minimum password length and collaborating with another intern to rate-limit password-reset requests, preventing spam abuse.
- Ran a full-site cross-browser QA pass, identifying and resolving 5+ CSS/UI bugs to improve consistency across browsers.

Integration Manager

Cypress, TX

NASA Moonshot Program

June 2025 - June 2025

- Led a 5-member team to create a preliminary design review for Artemis spacesuit technologies, supporting evaluation of design concepts and integration needs.
- Partnered with NASA engineers to develop CAD models and assess design tradeoffs, delivering a final design that improved subsystem integration and met all requirements.
- Delivered a 14-minute technical presentation scoring 242/280, demonstrating aerospace design expertise, rapid prototyping, and systems engineering.

Projects

Basic Rocket Flight Simulator

[GitHub](#) 

- Built and validated a multi-stage rocket flight simulator against closed-form analytical solutions within 1% on powered ascent and half a meter on the coast phase; streamed live telemetry via FastAPI/WebSockets to a Three.js 3D visualization.
- Tools: C++, Python, FastAPI, WebSockets, JavaScript, Three.js

ML/AI Stock Market Predictor

[GitHub](#) 

- Built LSTM pipeline incorporating RSI, MACD, and Bollinger Bands as engineered features to predict next-day equity direction; evaluated with RMSE and R^2
- Tools: Python, PyTorch, scikit-learn

Technical Skills

Tools: SOLIDWORKS, flow5, Git, VS Code, IntelliJ

Skill Sets: CAD, Composite Manufacturing, Dynamic Programming, Graph Algorithms, Data Structures

Languages: Java, Python, C++, PHP, SQL, JavaScript